

**Whole-body clearing of beetles by successive treatment of hydrogen peroxide and
CUBIC reagents: Whole-body clearing of beetles**

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18 **Abstract**

19 Internal tissues of multicellular organisms cannot directly be seen because they contain
20 pigments. For this reason, whole-body clearing methods have been developed and applied
21 to mammals such as mice. Insects such as beetles, however, cannot be cleared by the
22 mammalian method because of pigments such as melanin in their exoskeletons. In this
23 study, we tried to develop a whole-body clearing method for large beetles. We first
24 bleached the exoskeleton using a hydrogen peroxide treatment, and applied the the
25 advanced Clear, Unobstructed Brain/Body Imaging Cocktails and Computational analysis
26 (CUBIC) reagents to make the internal tissues transparent. The combined method of
27 hydrogen peroxide and the advanced CUBIC allowed us to successfully perform
28 whole-body clearing of the large beetles.

29

30 **Key words:** Whole-body clearing, hydrogen peroxide, CUBIC, *Trypoxylus dichotomus*,
31 *Rhomborrhina japonica*, *Dorcus titanus typhon*

32

INTRODUCTION

Mammalian tissues are opaque because they contain pigments that absorb visible light, such as hemoglobin, and myoglobin; substances that scatter light, such as lipids; and autofluorescent substances that interfere with the visualization of fluorescently labeled substances such as nicotinamide adenine dinucleotide phosphate hydrogen and collagen. Recently, whole-body clearing methods have been developed to make mammalian tissues transparent (Tainaka *et al*, 2016). However, these methods do not make insects transparent because of pigments such as melanin in their exoskeletons (Noh *et al*, 2016). Hydrogen peroxide has been used to bleach insect exoskeletons and make them transparent (Koga *et al*, 2009). Recently, it has been reported that bleaching the exoskeleton with hydrogen peroxide and clearing of internal tissues with Clear, Unobstructed Brain/Body Imaging Cocktails and Computational analysis (CUBIC) to successfully make small crustaceans transparent (Konno and Okazaki, 2018). In this study, we applied clearing methods such as hydrogen peroxide (Koga *et al*, 2009; Konno and Okazaki, 2018) and the advanced CUBIC (Susaki *et al*, 2015; Konno and Okazaki, 2018) to large beetles such as *rhinoceros* beetles; however, none of the methods were successful when used individually. However, by combining methods of hydrogen peroxide and the advanced CUBIC, here we were able to develop the whole-body clearing methods of large beetles.

MATERIALS

Animals

Large beetles, such as *Trypoxylus dichotomus* and *Rhomborrhina japonica*, were collected in Otomeyama Park (Shinjuku-ku, Tokyo, Japan). *Dorcus titanus typhon* was purchased from Konchukan in Toshimaen (Nerima-ku, Tokyo, Japan). They were stored as dry specimens (Figure 1).

Reagents

10% Formaldehyde Neutral Buffer Solution (Nacalai Tesque, cat. no. 37152-51)

4% Paraformaldehyde Phosphate Buffer Solution (PFA) (FUJIFILM Wako Pure Chemical Co. cat. no. 163-20145)

Ethanol (FUJIFILM Wako Pure Chemical Co. cat. no. 057-00451)

Hydrogen Peroxide (H₂O₂) (FUJIFILM Wako Pure Chemical Co. cat. no. 081-04215)

Urea (FUJIFILM Wako Pure Chemical Co. cat. no. 219-00175)

N,N,N',N'-tetrakis (2-hydroxypropyl) ethylenediamine (Quadrol) (Tokyo Chemical Industry, cat. no. T0781)

Triton X-100 (Sigma-Aldrich, cat. no. X100-500ML)

69 Nitrilotriethanol (FUJIFILM Wako Pure Chemical Co. cat. no. 145-05605)

70 Sucrose (FUJIFILM Wako Pure Chemical Co. cat. no. 196-00015)

71

72 **METHODS and RESULTS**

73 The method we developed for whole-body clearing of large beetles consisted of three
74 steps: I. Fixation, II. Bleaching of exoskeleton by hydrogen peroxide and III. Whole-body
75 clearing by the advanced CUBIC.

76 Step I. Fixation. We rehydrated the dry specimen of *Dorcus titanus typhon*, *Trypoxylus*
77 *dichotomus*, and *Rhomborrhina japonica* in phosphate buffered saline (PBS) (pH 7.4).
78 We prepared alcoholic formalin by diluting 10% formaldehyde neutral buffer solution
79 with three volumes of ethanol (Koga et al, 2009). We fixed them with alcoholic formalin
80 at 37°C for a week, and washed them with PBS three times.

81 Step II. Bleaching of exoskeleton by hydrogen peroxide. We incubated the fixed
82 samples with 6% H₂O₂/ethanol (30% H₂O₂ diluted with four volumes of ethanol) at 37°C
83 for 14 weeks (Figure 2) (Koga et al, 2009), replacing the 6% H₂O₂/ethanol twice per
84 week. Note that do not fix the cap tightly because bubbles are formed. After the bleaching,
85 we washed the bleached samples with PBS three times.

86 Step III . Whole-body clearing by the advanced CUBIC. We performed the whole-body
87 clearing according to the advanced CUBIC (Susaki et al, 2015). We prepared reagent 1,
88 containing 25%(w/w) Urea, 25%(w/w) Quadrol, and 15%(w/w) Triton X-100. We
89 incubated the bleached samples with 1/2 reagent 1 (reagent 1 diluted with an equal
90 volume of water) at 37°C for 2 days. We then replaced 1/2 reagent 1 to reagent 1. We
91 exchanged reagent 1 twice per week, and incubated for 5 weeks (Figure 3).

92 We then prepared reagent 2, containing 25%(w/w) Urea, 50%(w/w) Sucrose, 10%(w/w)
93 nitrilotriethanol. We incubated the reagent 1-treated samples with 1/2 reagent 2 (reagent 2
94 diluted with an equal volume of PBS) at 37°C for 1 day. We then replaced 1/2 reagent 2 to
95 reagent 2. We exchanged reagent 2 every week, and incubated for 2 weeks (Figure 4).
96 Washing the reagent 2-treated samples with PBS decreased the transparency, so we kept
97 the final samples in the reagent 2.

98 We compared our method with other methods (Table, Figure 5). Only our method, but
99 not other methods, succeeded in making large beetles transparent. The differences
100 between our methods and others are the combination of hydrogen peroxide, which
101 bleaches exoskeletons, and CUBIC, which clears both bleached exoskeletons and internal
102 tissues, with much longer incubation time. The combination of hydrogen peroxide and
103 CUBIC in this study was the same as the method used by Konno and Okazaki, however,

our method used a longer incubation time with both hydrogen peroxide and CUBIC, which enabled the whole-body clearing of the beetles.

DISCUSSION

In this study, we developed a method of whole-body clearing of large beetles which combined bleaching exoskeletons using hydrogen peroxide with whole-body clearing using CUBIC. During the bleaching process, when ethanol was used as a solvent (Koga *et al*, 2009), the internal tissue was sufficiently fixed during the whole procedure, but when water was used as a solvent (Konno and Okazaki, 2018), the internal tissue was not sufficiently fixed (data not shown). Therefore, ethanol should be used as the solvent during bleaching with hydrogen peroxide. Furthermore, while our study bleached the large beetles for 14 weeks, this did not completely bleach them. A longer bleaching duration may be required to completely bleach exoskeletons. The whole-body clearing by CUBIC required about 3 months, which is longer than required for clearing mouse tissues (about 2 weeks) (Susaki *et al*, 2015). The use of the latest CUBIC reagents (Tainaka *et al*, 2018) may shorten this long-term treatment by the current protocol. We did not perform nuclear staining, and did not visualize the internal tissues. It remains unclear whether the internal tissues are correctly fixed.

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146 Figure legends

147

148 **Figure 1** Dry specimens of *Dorcus titanus typhon* (left), *Trypoxylus dichotomus* (middle),
149 and *Rhomborrhina japonica* (right). A grid, 1 centimeter.

150

151 **Figure 2** Samples incubated with 6% H₂O₂/ethanol for 1 week (top), 8 weeks (middle),
152 and 14weeks (bottom). *Rhomborrhina japonica* (left), *Trypoxylus dichotomus* (middle),
153 *Dorcus titanus typhon* (right).

154

155 **Figure 3** Samples treated with reagent 1 of CUBIC for 5 weeks. *Rhomborrhina japonica*
156 (left), *Trypoxylus dichotomus* (middle), *Dorcus titanus typhon* (right).

157

158 **Figure 4** Samples treated with reagent 2 of CUBIC for 2 weeks. The final samples of the
159 whole-body clearing. *Rhomborrhina japonica* (left), *Trypoxylus dichotomus* (middle),
160 *Dorcus titanus typhon* (right).

161

162 **Figure 5** Samples treated with the methods by Susaki 2015, Konno 2018, Koga 2009,
163 and this study (from left to right).

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Table. Comparison with other methods.

	Fixation	Fixation Duration	Bleaching	Bleaching Duration	Clearing	Clearing Duration
Susaki <i>et al</i> , 2015	PFA	overnight	none	none	CUBIC	10 days
Konno and Okazaki, 2018	PFA	overnight	H ₂ O ₂	2 days	CUBIC	2 days
Koga <i>et al</i> , 2009	Formalin/Ethanol	1 week	H ₂ O ₂ /Ethanol	3 weeks	none	none
This study	Formalin/Ethanol	1 week	H ₂ O ₂ /Ethanol	14 weeks	CUBIC	7 weeks

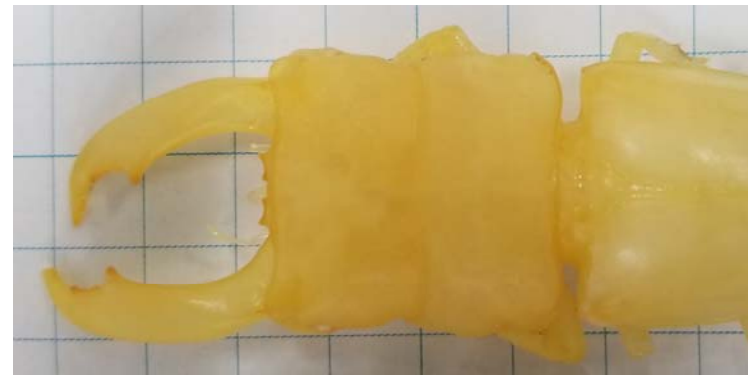
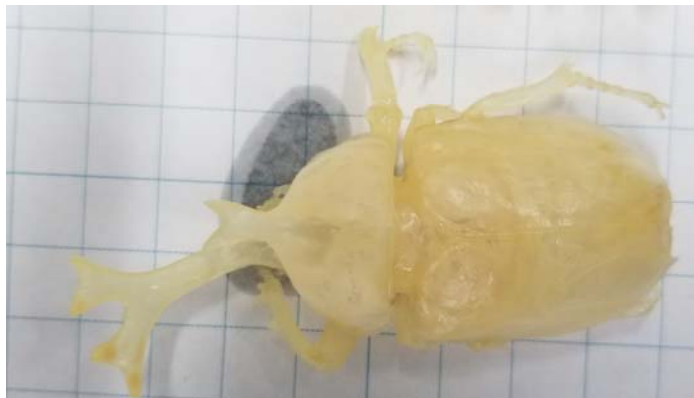


Figure 1

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Figure



Figure 3



Figure 4



Figure 5